WEEK 1 · SESSION L2 · UNIT I

Why HR Is Going Digital

Industry 4.0, artificial intelligence and hybrid work — and what the three of them together are doing to the HR function and to the nature of work itself.

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Built around Project Disha — the GreenLeaf Retail digital-HR transformation simulation

SESSION MAP

60 min

The hour ahead

0:00

Where you already live with this

Ice-breaker + opening
Socratic dialogue

0:15

The three forces

Industry 4.0 · AI ·
Hybrid and platform work

0:35

Digitization vs transformation

The distinction the whole course turns on

0:50

GreenLeaf, and what comes next

Current state, the four anchors, bridge to R1

Leave with this: one paragraph you could say to a skeptical CFO explaining why GreenLeaf must transform its people systems — not merely computerize them.

0:00 – 0:10 · EVERYONE PARTICIPATES

10 min

Ice-breaker: “Paper on a Screen”

1

Write one line

2 min

A digital process you personally used in the last month that annoyed you. University portal, exam form, bank KYC, ticket booking, a delivery app. One line only.

2

Swap and judge

3 min

Hand it to the person beside you. They decide: was this DIGITIZED (paper moved onto a screen) or TRANSFORMED (the process itself redesigned)? You may not defend yours.

3

Put it on the grid

4 min

Come up and place the slip on the board grid: effort saved (low → high) against experience (worse → better).

4

Read the cluster

1 min

We look at where the slips bunched — and name what we just did.



You have just run a digital-maturity diagnostic on your own life. In two weeks you will run one on an 8,000-person retailer.

OPENING SOCRATIC DIALOGUE · 0:10 – 0:15

“What actually changes when HR goes digital?”

Who

gets faster — and who quietly gets slower?

What

decision moves from a person to a rule?

Whose

judgment is now encoded, and can they still see it?

Hold the question open. Three answers, no consensus, then we go looking for the forces underneath it.

20 min

Three forces, arriving together

I Industry 4.0

Cyber-physical systems, IoT, cloud and connectivity collapse the distance between the physical shop floor and the data layer.

Work becomes measurable in real time — including yours.

II Artificial Intelligence

Prediction becomes cheap. Screening, matching, ranking and forecasting move from human judgment to model output.

Decisions become fast, scalable — and harder to explain.

III Hybrid & Platform Work

Where work happens, and the contract under which it happens, both come loose from the office and the payroll.

The workforce stops being one population.

None of these is an HR initiative. All three land on HR's desk.

That is the whole argument for why the HR function must change shape — it did not choose any of this, and it cannot opt out of any of it.

SCHWAB (2016)

0:15

Force I — the fourth industrial revolution



Steam & mechanization

HR did not exist. Labor was hired at the gate.

Electricity & mass production

Personnel administration appears — records, payroll, discipline.

Electronics & IT

HRM professionalizes. Systems of record arrive: the HRIS.

Cyber-physical, AI, data

Every process emits data. HR is asked to decide with it — and to answer for it.

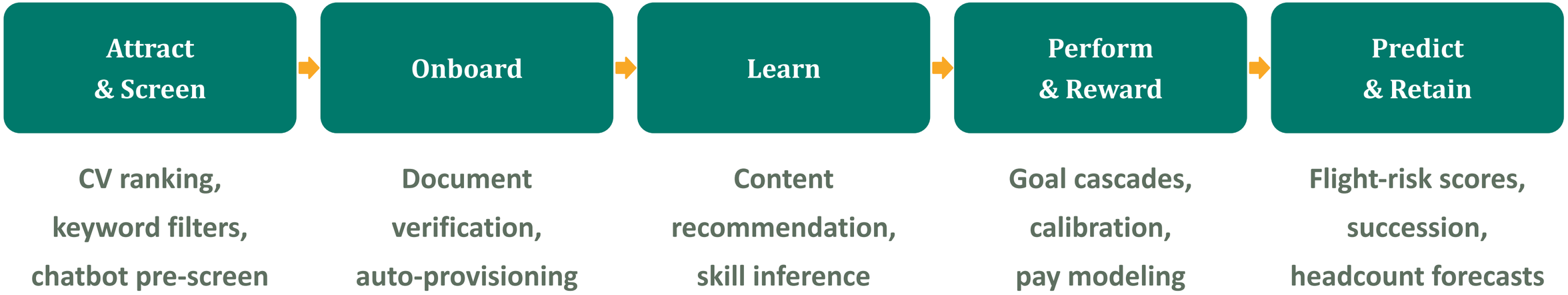
Each previous revolution changed what HR administered. This one changes what HR is asked to know.

What Industry 4.0 does to the work itself

Tasks split from jobs	Roles stop being stable bundles. A job becomes a shifting portfolio of tasks, some of which leave for a machine each year.
Skills develop a half-life	The useful life of a technical skill shortens. Learning stops being an event and becomes infrastructure.
Work emits data continuously	Every login, scan, route and click is recorded. Measurement arrives whether or not anyone designed it.
The firm loses its edges	Employees, contractors, apprentices and platform partners work side by side under four different contracts.

Every one of these four lands on an HR process before it lands anywhere else.

Force II — where the algorithm already sits



What genuinely improves

- Speed and consistency at volume
- Patterns no manager could see unaided
- Administrative load lifted off HR staff

What quietly gets worse

- Past bias is learned, scaled and made routine
- The reason for a decision becomes unavailable
- Accountability diffuses — nobody decided, the model did

GRATTON (2021)

0:27

Force III — hybrid work, and the equity trap

Jobs & tasks

Which tasks actually need co-location, concentration, coordination or cooperation?

Employee preferences

Autonomy over place and time is now a retention variable, not a perk.

Projects & workflows

Redesign the workflow, or hybrid simply degrades it.

Inclusion & fairness

Who is eligible for flexibility — and who is structurally not?

The trap

A store associate cannot work from home. A warehouse operator cannot flex her hours. A delivery partner has no office to return to.

If flexibility is designed only for the people who already have laptops, hybrid does not liberate the workforce — it divides it in two, and HR built the wall.

At GreenLeaf: roughly 180 stores, three distribution centers and a growing last-mile partner network. The desk-based, English-fluent, laptop-owning employee is a **minority** of the workforce — and the default user of almost every HR product on the market.

The Indian layer — why imported frameworks slip

The divide is inside the firm

Corporate HQ is moderately digitized; the store floor and warehouse are largely manual. Vernacular language, shared devices and thin bandwidth are the norm, not the exception.

Platform work is scaling fast

NITI Aayog (2022) estimates India's gig workforce at roughly 7.7 million in 2020–21, rising towards 23.5 million by 2029–30 — people managed by an algorithm, outside the payroll.

The law has arrived

The Digital Personal Data Protection Act, 2023 governs employee data and consent. The Code on Social Security, 2020 and state gig-worker legislation begin to reach platform labor.

Public rails change the design

India Stack, DigiLocker, Aadhaar-based verification and UPI make onboarding flows possible here that most global HR vendors never built for — with privacy trade-offs of their own.

A platform designed in California rarely lands cleanly in a Tier-3 town.

Where the three forces meet: four shifts in HR

01 From records to evidence

HR is asked to prove effects, not report activity. 'We ran 40 sessions' stops being an answer.

02 From process owner to product owner

HR services are now interfaces people either use or abandon. Adoption becomes an HR metric.

03 From administrator to governor

Someone must own algorithmic fairness, consent and data residency. In practice, that is HR.

04 From employee relations to workforce relations

Employees, contractors, apprentices and platform partners — four populations, one duty of care.

This is the job description you are being trained for. It is not the one in most textbooks.

Digitization is not transformation

DIGITIZATION

The same process, on a screen

- The form becomes a PDF
- The register becomes a spreadsheet
- The queue becomes an inbox
- The old rule is now automatic — including the parts that were wrong
- Cost falls a little; experience rarely improves

TRANSFORMATION

A different process, because the constraint moved

- The question is re-asked: why does this step exist?
- The decision right moves to where the data is
- The service reaches people it never reached
- Roles are redesigned, not merely re-tooled
- The measure of success changes

Most “digital HR” programs are digitization projects wearing a transformation budget.

5 min

Now classify these five

1. The annual paper engagement survey is reissued as a PDF form emailed to all staff.
2. Leave applications move from a register to a shared spreadsheet that HR updates each evening.
3. Store associates get a vernacular mobile app that shows their leave balance and approves leave instantly.
4. The recruitment inbox is replaced by an applicant-tracking system that ranks CVs using the same rule the hiring manager always used.
5. Payroll queries are answered by a chatbot, and the three payroll clerks are retrained as data-quality owners.

D

or

T

Mark each one D or T.

Then defend the one your partner marked differently.

One of these five is the hardest to call. Find it.

7 min

Your client: GreenLeaf Retail

8,000

employees across HQ,
4 regional offices,
~180 stores, 3 DCs

38%

annual attrition
in stores

47

days to hire a
frontline role

41%

response rate on the
annual paper survey

What the CHRO inherited

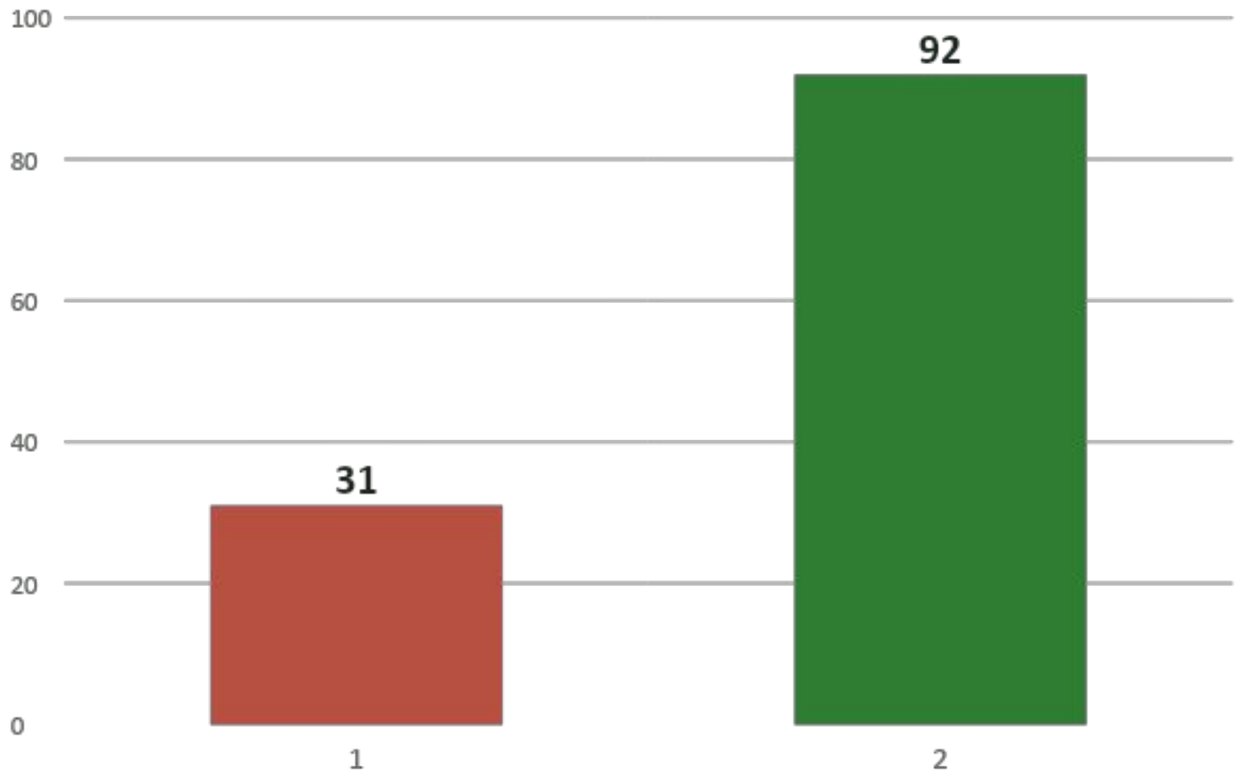
- Core employee records in three incompatible spreadsheets and a decade-old on-premise HRIS module
- Recruitment run from email and a shared inbox
- Learning delivered as classroom sessions and PDFs
- Engagement measured once a year, on paper
- No analytics function to speak of

The Board's question

"Why does a digital-first retailer still run its workforce on paper?"

A new CHRO has been appointed. She has commissioned a Digital HR Transformation Office and staffed it with you. From Week 2 you own the answer.

The gap that breaks most rollouts



Read it as a design brief

Any HR system that assumes a company email address reaches roughly a third of this workforce.

Any system that assumes a personal smartphone reaches almost all of it — if it is mobile-first, low-bandwidth and available in the language people actually speak.

That single sentence will decide more about your transformation's success than any vendor comparison you run in Round 2.

INTRODUCED PROPERLY IN L3 — MEET THEM NOW

Four people your design must work for

GL-1001

LN Lakshmi Naidu

Tier-3 store associate. Smartphone only. Prefers Telugu. Low digital literacy.

Tests: the digital divide

GL-1002

AJ Arjun Joshi

Bronze-tier delivery partner, Chennai. App-managed, paid per delivery. Not an employee.

Tests: algorithmic fairness

GL-1003

SM Sunil Mehta

HR Officer, Lucknow. Seventeen years' tenure. Anxious that automation will make him redundant.

Tests: change and reskilling

GL-1004

SI Shreya Iyer

E-commerce executive, Bengaluru. Digitally fluent, frustrated by the legacy stack and the absence of data.

Tests: enablement at the edge

Every decision your pod makes for the next twelve weeks gets stress-tested against all four. A design that serves only Shreya is not a transformation.

0:57 – 1:00 · BRIDGE TO NEXT WEEK

3 min

From “why” to “how mature”

1	**Today**	Why the forces make transformation unavoidable
2	**L3, this week**	GreenLeaf's current state in detail; pods log in to Disha
3	**Week 2**	Drivers and maturity models — Westerman's digital-masters grid
4	**Week 2, L3**	Round 1 launches: diagnose GreenLeaf's maturity and choose where to start

Before Week 2

- Westerman, Bonnet & McAfee (2014), Leading Digital — Chapters 1–2
- Scully-Russ (2019), Industry 4.0 and HRD
- Deloitte, Global Human Capital Trends — selected sections
- Quiz 1 opens Week 2, L1 — twelve minutes, closed book

Forum prompt: name one “digital” HR process you have met that was really just paper on a screen. What was missing?

THREE THINGS TO CARRY OUT OF THIS ROOM

1

Industry 4.0, AI and hybrid work are not three HR projects.

They are three external forces that arrived together and landed on HR's desk. HR did not choose them and cannot opt out of them.

2

Digitization moves a process onto a screen. Transformation asks why the process exists.

Most programs that call themselves the second are quietly the first — and the budget line is the tell.

3

Reach is the constraint, not features.

A design that works only for the desk-based, English-fluent minority has not transformed the workforce. It has divided it.

Next: L3 — Disha briefing, the GreenLeaf current-state pack, and your first login.

Going deeper: the theory underneath the digital shift

The next slides set out Zuboff, the task model of Autor, Levy and Murnane, Orlikowski, and the three stages of digital change: what each claims, how it works, where it breaks, and what it means at GreenLeaf.

THEORY IN DEPTH · ZUBOFF (1988)

Technology can automate, or it can informate

THE CORE IDEA

Automating replaces human effort. Informating produces new information about the work itself.

HOW IT WORKS

The same system does both. The choice is whether the new information empowers workers or only watches them.

LIMITS AND CRITIQUE

Written before the internet; later Zuboff (2019) argued informating turned into surveillance for profit.

AT GREENLEAF

An ESS app automates leave requests. It also informates: every login now tells HR who uses it and who does not.

THEORY IN DEPTH · AUTOR, LEVY AND MURNANE (2003)

Computers replace routine tasks, not whole jobs

THE CORE IDEA

Jobs are bundles of tasks. Machines substitute for routine tasks and complement non-routine ones.

HOW IT WORKS

Routine cognitive tasks (record keeping) and routine manual tasks go first; judgment and personal service resist.

LIMITS AND CRITIQUE

AI now reaches some non-routine cognitive tasks, so the routine line is moving.

AT GREENLEAF

Sunil Mehta's data entry is routine; his process knowledge and judgment are not. Redesign the job, do not delete it.

THEORY IN DEPTH · ORLIKOWSKI (1992)

Technology shapes people, and people reshape technology

THE CORE IDEA

The duality of technology: systems are designed by people, then constrain them, and are reinterpreted in use.

HOW IT WORKS

Users adapt, work around, or ignore features. The system that runs is not the system that was bought.

LIMITS AND CRITIQUE

The idea explains a lot after the fact but predicts little in advance.

AT GREENLEAF

Spreadsheet Trackers at adoption 1.00 are a workaround users built because the official systems did not fit their work.

THEORY IN DEPTH · VERHOEF ET AL. (2021)

Three stages: digitization, digitalization, transformation

THE CORE IDEA

Digitization turns analog into digital. Digitalization changes processes. Transformation changes the business model.

HOW IT WORKS

Each stage needs different resources: tools first, then process redesign, then new capabilities and structures.

LIMITS AND CRITIQUE

Real firms sit in different stages in different functions at once; the stages are a lens, not a ladder.

AT GREENLEAF

GreenLeaf's PDF survey is digitization; a vernacular mobile leave flow is digitalization; HR as a data-led partner is transformation.